18 Energy Balances for CSTRs

Why are we learning this?

In the previous lecture, we developed the General Energy Balance Equation. Today we will apply that equation to a specific reactor type: the **Continuous Stirred Tank Reactor (CSTR)**. Unlike the isothermal reactors studied earlier, non-isothermal reactors introduce an important complication:

Temperature affects reaction rate. Reaction affects temperature.

As a result, conversion and temperature must be solved simultaneously. By the end of today's lesson, you should understand:

- How the CSTR energy balance is derived.
- Why heat transfer appears in the energy balance.
- Why shaft work is usually neglected.
- The difference between sensible heat and heat of reaction.
- Why non-isothermal reactor problems require both a mole balance and an energy balance.
- How conversion and temperature become coupled unknowns.

Course Roadmap

Introduction to CRE
↓
General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
Reaction Progress
(Conversion)
↓
Graphical Reactor Design

(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)
↓
Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
►Energy Balances◄
↓
One Reaction: Non-Isothermal Reactor Design
↓
Multiple Reactions
↓
Reactor Safety
↓
Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

Mole Balance
↓
Predicts Conversion

Energy Balance
↓
Predicts Temperature

Both equations are needed to describe a non-isothermal reactor.

What Changes in Non-Isothermal Reactor Design?

In previous reactor design problems, temperature was usually known. Now temperature becomes an unknown. This creates an important feedback loop:

Higher Temperature
↓
Faster Reaction Rate
↓
Different Conversion
↓
Different Heat Release
↓
Different Temperature

This coupling is the central challenge of non-isothermal reactor design.

Let's derive the energy balance for a CSTR:

Starting from the general form (assuming steady state):

In the CSTR, the stirrer is doing work **on** the reactor fluid and hence increasing its energy. However, this amount of work is usually very small relative to the heat of reaction. So, we can usually set $\dot{W}_s = 0$. Exception: If the fluid is viscous and you are given a value for $\dot{W}_s$.

Why Is Shaft Work Usually Negligible?

The agitator in a CSTR adds energy to the fluid. However, chemical reactions often release or absorb much more energy than the stirrer contributes.

For most reactor-design problems: Heat of Reaction >> Shaft Work

As a result, shaft work is commonly neglected unless a problem explicitly indicates otherwise.

$\dot{Q}$ term: Two cases:

- $T_{a1} \neq T_{a2}$: Rate of heat transfer varies over the heat transfer area. In this case, use
- $T_{a1} = T_{a2}$: Rate of heat transfer is constant over the heat transfer area. In this case, use

For the purposes of this derivation, let's use the second case. Then:

Why Does Heat Transfer Matter?

Reactors exchange energy with their surroundings.

Heat transfer may:

Remove Heat

for exothermic reactions

or

Add Heat

for endothermic reactions.

The heat-transfer term allows the reactor temperature to be controlled and is often critical for safe operation.

In the case of just one reaction, F terms can be related to F_0 terms using X:

Expanding terms:

Further simplify by using the definition that $\Theta_i = \frac{F_{i0}}{F_{A0}}$:

Grouping like terms:

Where $\Delta H^{rxn} = \frac{d}{a}H_D + \frac{c}{a}H_C - \frac{b}{a}H_B - H_A$. Therefore:

$H_{i0} - H_i$ terms are **sensible heats**, equal to $C_P(T_0 - T)$. Substituting in:

Heat of Reaction and Sensible Heat Are Different

Heat of Reaction

Energy associated with: Breaking Bonds + Forming Bonds

Sensible Heat

Energy associated with: Changing Temperature

Both contributions appear in reactor energy balances, but they describe different physical phenomena.

Solving for *X*:

$$X_{EB} = \frac{F_{A0}C_{PA}(T_0 - T) + F_{A0}C_{PB}\Theta_B(T_0 - T) + UA(T_a - T)}{F_{A0}(\Delta H^{rxn})}.$$

Conversion Helps Simplify the Energy Balance

Just as in mole balances, reaction stoichiometry allows flow rates to be written in terms of conversion.

This means the energy balance can also be rewritten in terms of:

Conversion
+
Temperature

rather than tracking every species individually.

Ultimately, we have 2 unknowns, *X* and *T*, hence, we need 2 equations: a mole balance and an energy balance. We denote this equation EB to denote that it is the energy balance.

The mole balance is: $V = \frac{F_{A0}X_{MB}}{-r_A}$, where $-r_A = kC_AC_B^{\frac{b}{a}},\ k = k_1(T_1)\exp\left[\frac{E}{R}\left(\frac{1}{T_1} - \frac{1}{T}\right)\right]$, and $C_i = \frac{C_{A0}\left(\Theta_i - \frac{\nu_i}{\nu_{LR}}X\right)}{1+\epsilon X}\frac{T_0}{T}$.

The mole balance can be solved for *X*, giving a function $X_{MB}(T)$.

Temperature Appears Everywhere

Temperature influences:

- Rate constants
- Concentrations
- Heat-transfer behavior
- Equilibrium (in many systems)

This is why temperature can dramatically alter reactor performance.

We now have a system of 2 equations and 2 unknowns, which can be solved by hand, with mathematical software, or graphically.

Why One Equation Is No Longer Enough

For isothermal reactor design:

Unknown = Conversion

One equation was often sufficient.

For non-isothermal reactor design:

Unknowns:
Conversion
Temperature

Therefore, we need:

Mole Balance
+
Energy Balance

to fully describe the reactor.

How Do We Solve Non-Isothermal CSTR Problems?

The general strategy is:

1. Write the mole balance.
2. Write the energy balance.
3. Express both in terms of X and T.
4. Solve the two equations simultaneously.

Solutions may be obtained:

- Graphically
- By hand (when simple enough)
- Using mathematical software

Summary and Key Takeaways

Today we derived the CSTR energy balance and connected it to reactor design.

Key Ideas

☑ Non-isothermal reactor design requires both a mole balance and an energy balance.

☑ Heat transfer and reaction heat both influence reactor temperature.

☑ Shaft work is usually negligible relative to the heat released or absorbed by reaction.

☑ Sensible heat accounts for temperature changes, while heats of reaction account for energy stored in chemical bonds.

☑ Conversion and temperature become coupled unknowns.

☑ Two equations are required because two unknowns must be solved for simultaneously.

Most Important Idea

Conversion affects Temperature. Temperature affects Conversion. The CSTR energy balance captures this interaction, making it possible to predict the behavior of non-isothermal reactors.